Analyzing Data with EXCEL

Examining the Availability of Hospital Beds for Cancer Treatment in NHS Scotland.

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Introduction:

The purpose of this research is to give an analysis of approaches for anticipating and optimising hospital bed availability in NHS Scotland for cancer treatment. It covers the difficulties that the NHS is experiencing because of the COVID-19 epidemic, the growing backlog of cancer treatments, and the necessity for good planning and resource allocation. The report showcases how forecasting, linear programming, and business intelligence assist NHS Scotland in managing bed availability.

Forecasting Evaluation:

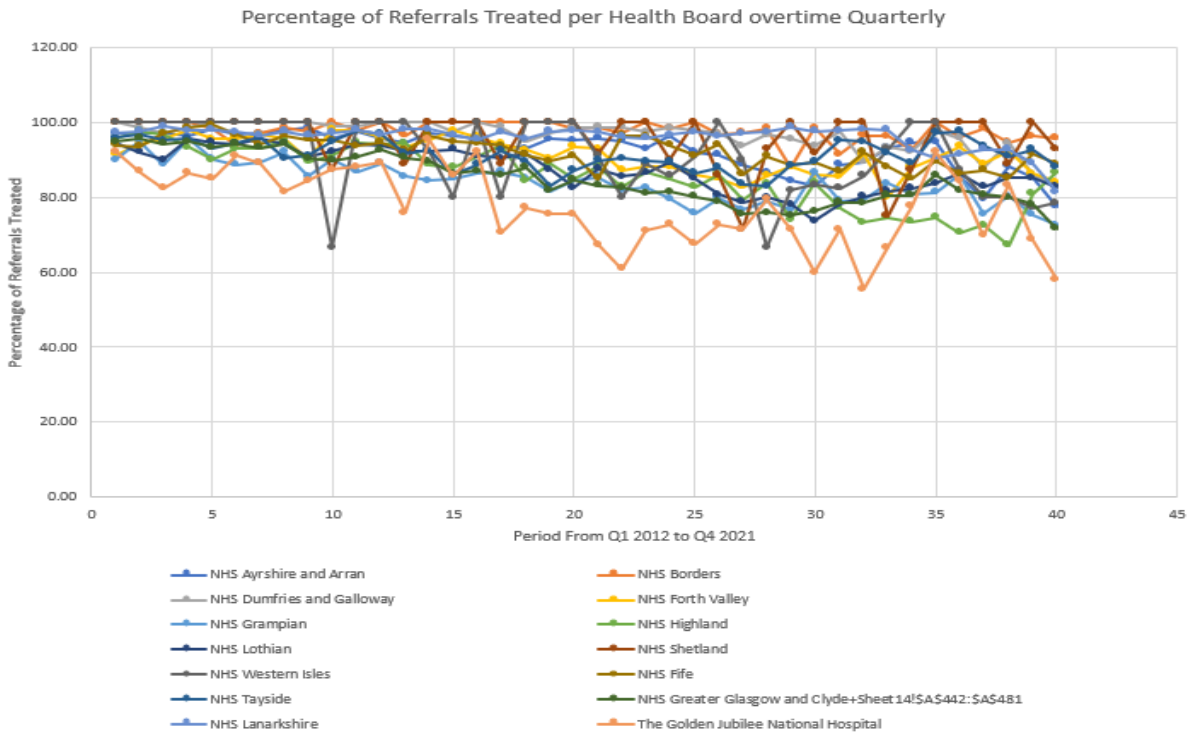
Management at NHS Scotland should consider four key questions to measure the success of their forecasting techniques: accuracy in predicting future demand, timeliness for prompt decision-making, flexibility to adapt to changes, and usability for healthcare professionals. Challenges in applying forecasting processes include data quality, uncertainty in demand forecasting, resistance to change, resource allocation, and ethical considerations. These challenges can impact forecast accuracy, decision-making, and resource allocation. Overcoming them requires collaboration, continuous evaluation, and addressing ethical concerns. By addressing these questions and challenges, NHS Scotland can enhance forecasting techniques, improving healthcare planning and resource allocation.

Data Access, Cleaning and Preparation:

a) Cancer Waiting Time Dataset:

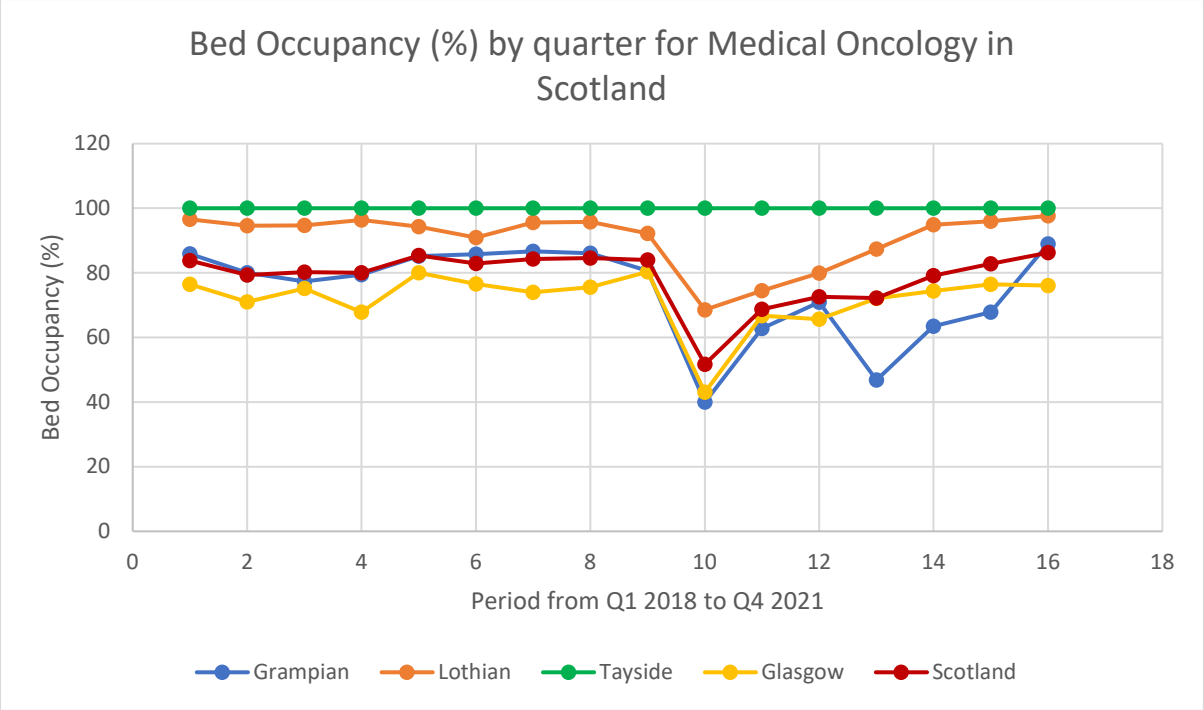
Several trends and patterns emerge when plotting the percentage of cancer referrals that have gotten treatment over time. Grampian has failed to meet the aim of 90% initial treatment received by the patient within 62 days. The aim was met by NHS Borders, Dumfries and Lanarkshire health board. Golden Jubilee did not meet its goal from 2016. And, while the remaining health boards have met the target, there is a drop around 2020 owing to the impact of COVID-19. Variations in proportion across health boards over time could be related to factors such as healthcare resource distribution, personnel levels, infrastructure, or geographical variances in patient demographics. Further research could include looking into the causes of the observed trends. Furthermore, it would be beneficial to compare the performance of different health boards in order to uncover potential best practises. It is recommended to investigate qualitative data, such as patient opinions and experiences, to acquire a more comprehensive understanding. The health boards are confronting a number of issues, including three cancellations of patient surgeries owing to ICU shortages and low waiting time performance. Forecasting the following year may aid in meeting the targets in resolving these difficulties. Collaboration among healthcare professionals, policymakers, and stakeholders is vital. Together, they can develop interventions to reduce waiting times, improve patient pathways, and increase public awareness of early detection and treatment.

These efforts can enhance timely treatment for cancer referrals, leading to improved patient outcomes and experiences.



b) Beds by Treatment and Specialty Board

The given dataset consists of the bed occupancy rate of all the departments in hospitals. But here we need to analyse only the bed occupancy rate of the Medical Oncology department. It is the department which deals with the treatment & diagnosis of tumours. So, for that first we must clean and prepare data. We will filter the data of Medical Oncology department using the Filter option. After the analysis, we can understand that the bed occupancy rate of all the three health boards other than Tayside has a large decrease at the time period of 2020 because of the Covid issues. The reasons for this can be the less transportation at the time of Covid or the fear of people to get to the Hospitals. And the other thing we can understand is Tayside is having full bed occupancy all the time that can be because of their food, water or lifestyle which causes in most of the people. From the below graph, we can observe all these findings.



Time Series Forecasting:

We examined the influence of Covid in the medical oncology department for three health boards- Grampian, Lothian, and Tayside- with and without the year 2020 for time series forecasting. We accomplished this by employing three methods: linear trend, forecast with seasonality, and exponential smoothing. The graphic below depicts the mean squared error for the year 2020:

		With 2020		
Dataset	Health Board			
		Forecast Method	MSE	
Cancer Referrals	NHS Grampian	Linear Trend	9.34	
Cancer Referrals	NHS Grampian	Linear Trend with Seasonality	8.15	Chosen Forecast Method
Cancer Referrals	NHS Grampian	Exponential Smoothing	12.04	
Cancer Referrals	NHS Lothian	Linear Trend	11.82	
Cancer Referrals	NHS Lothian	Linear Trend with Seasonality	11.54	
Cancer Referrals	NHS Lothian	Exponential Smoothing	6.91	Chosen Forecast Method
Cancer Referrals	NHS Tayside	Linear Trend	14.89	
Cancer Referrals	NHS Tayside	Linear Trend with Seasonality	14.87	
Cancer Referrals	NHS Tayside	Exponential Smoothing	11.58	Chosen Forecast Method
		With 2020		
Dataset	Health Board			
		Forecast Method	MSE	
Bed occupancy	NHS Lothian	Linear Trend	66.56	
Bed occupancy	NHS Lothian	Linear Trend with Seasonality	61.3	
Bed occupancy	NHS Lothian	Exponential Smoothing	54.33	Chosen Forecast Method
Bed occupancy	NHS Grampian	Linear Trend	163.23	
Bed occupancy	NHS Grampian	Linear Trend with Seasonality	126.17	Chosen Forecast Method
Bed occupancy	NHS Grampian	Exponential Smoothing	213.44	
Bed occupancy	NHS Tayside	Linear Trend	0	
Bed occupancy	NHS Tayside	Linear Trend with Seasonality	0	
Bed occupancy	NHS Tayside	Exponential Smoothing	0	

Linearity Trend with Seasonality was chosen as the best forecast method for Grampian Health Board of both datasets since it had the lowest MSE. For Lothian, the best forecast method for both datasets is Exponential Smoothing. For Tayside, the best forecast method for the Cancer dataset is Exponential Smoothing. In the instance of the Bed dataset, Tayside's bed occupancy is 100% for the entire period, hence creating forecast methods is unnecessary because the output will be 0.

The figure below shows the analysis result without 2020.

		Without 2020		
Health Board	Datasets			
		Forecast Method	MSE	
Cancer Referrals	NHS Grampian	Linear Trend	6.69	
Cancer Referrals	NHS Grampian	Linear Trend with Seasonality	5.23	Chosen forecast method
Cancer Referrals	NHS Grampian	Exponential Smoothing	10.79	
Cancer Referrals	NHS Lothian	Linear Trend	10.35	
Cancer Referrals	NHS Lothian	Linear Trend with Seasonality	10.16	
Cancer Referrals	NHS Lothian	Exponential Smoothing	7.32	Chosen forecast method
Cancer Referrals	NHS Tayside	Linear Trend	12.37	
Cancer Referrals	NHS Tayside	Linear Trend with Seasonality	12.67	
Cancer Referrals	NHS Tayside	Exponential Smoothing	10.37	Chosen forecast method
		Without 2020		
Dataset	Health Board			
		Forecast Method	MSE	
	NHS Grampian	Linear Trend	127.58	
	NHS Grampian	Linear Trend with Seasonality	92.02	Chosen forecast method
	NHS Grampian	Exponential Smoothing	184.57	
	NHS Lothian	Linear Trend	7.21	
	NHS Lothian	Linear Trend with Seasonality	4.44	Chosen forecast method
	NHS Lothian	Exponential Smoothing	13.81	
	NHS Tayside	Linear Trend	0	
	NHS Tayside	Linear Trend with Seasonality	0	
	NHS Tayside	Exponential Smoothing	0	

The MSE is lower when forecasting the cancer dataset without including the data from 2020 due to the unique circumstances and disruptions caused by the COVID-19 pandemic. The pandemic led to significant changes in cancer care, including delays in screenings, rescheduling of treatments, and shifts in healthcare resources and priorities. Including the data from 2020 in the forecasting model might introduce anomalies and noise that do not reflect the usual patterns and trends of cancer cases. By excluding the 2020 data, the forecasting model can focus on the pre-pandemic period, capturing more accurate and consistent patterns, which results in a lower MSE.

Linear Optimization:

a) Optimization modelling optimizes resource allocation and timely treatment delivery in NHS Scotland. It informs bed availability, staff allocation, and patient scheduling for informed decision-making.

Potential Decision Variables:

1. The number of beds assigned to particular medical specialties.
2. Determining shift staffing.

3. Managing patient flow.

Result Variables:

1. Bed utilization.
2. Staffing efficiency: Measuring the workload distribution, ensuring quality care without excessive staffing.

Objective:

The objective of the optimization model can be to maximize bed utilization, optimize staffing, minimize patient wait times, or reduce costs while ensuring quality care.

Uncontrollable Variables:

1. Patient demand and arrival patterns.
2. Medical guidelines and regulations.
3. Staff availability and skills.

b) To create a linear programming model for staff allocation, define decision variables, objective function, and uncontrollable variable.

Decision Variables:

Let's represent the staff allocation for each 4-hour time period as:
S1: Number of staff allocated from 09:00-13:00
S2: Number of staff allocated from 13:00-17:00
S3: Number of staff allocated from 17:00-21:00
S4: Number of staff allocated from 21:00-01:00
S5: Number of staff allocated from 01:00-05:00
S6: Number of staff allocated from 05:00-09:00

Result Variable and Objective Function:

The objective is to minimize the total number of staff allocated for 24 hours.

Minimize: $Z = S1 + S2 + S3 + S4 + S5 + S6$

Uncontrollable Variables (Constraints):

The staff assigned to each time period should satisfy or exceed the corresponding demand.
$S1 + S2 + S3 >= 36$ (For Shift 1)
$S2 + S3 + S4 >= 26$ (For Shift 2)

S3 + S4 + S5 >= 22 (For Shift 3)
S4 + S5 + S6 >= 16 (For Shift 4)
S5 + S6 + S1 >= 10 (For Shift 5)
S6 + S1 + S2 >= 6 (For Shift 6)

All decision variables should be non-negative:

S1, S2, S3, S4, S5, S6 >= 0

This linear programming methodology tries to reduce the total number of personnel assigned while still meeting demand for each time period. You can get the optimal number of employees for any 24-hour period by solving this model.

Ethical Issues:

Legal Issues:

a. Data Protection Laws: NHS Scotland must comply with data protection regulations such as GDPR, ensuring lawful handling of personal data, including patient information.

b. Healthcare Regulations: Specific regulations govern patient confidentiality, consent, and handling of medical data in the healthcare sector.

c. Intellectual Property Rights: NHS Scotland must respect intellectual property rights and comply with licensing agreements when using proprietary software, algorithms, or data sources in their business intelligence systems.

Ethics Issues:

a. Privacy and Confidentiality: NHS Scotland must prioritize patient privacy and confidentiality, implementing robust security measures to protect sensitive data from unauthorized access, breaches, or misuse.

b. Fairness and Bias: Business intelligence systems should be designed to mitigate biases and ensure equal treatment of patients, addressing any potential discrimination or biases in algorithms or data collection.

c. Informed Consent: Obtaining informed consent from patients regarding data collection and usage in business intelligence systems is essential. Transparent communication about the purpose and scope of data usage builds and maintains patient trust.

Privacy Issues:

a. Data Minimization: NHS Scotland should practice data minimization by collecting and retaining only necessary data for business intelligence systems, reducing the risk of unauthorized access or breaches.

b. Anonymization and De-identification: Patient data should undergo appropriate anonymization or de-identification techniques to protect privacy and prevent individual identification.

c. Access Controls: Implementing strict access controls, including role-based access, strong authentication, and regular reviews, is crucial to restrict system access to authorized personnel and prevent unauthorized use or disclosure of patient data.

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